Targeting MEK in cancer and beyond: mechanistic insights and therapeutic opportunities

Wei Yen Chan, Ines Pires da Silva, Georgina V Long, Helen Rizos

MEK inhibitors are established therapies in *BRAF*-driven cancers, yet their broader clinical effect is limited by toxicity, resistance, and modest durability as a monotherapy, particularly in *RAS*-mutant tumours. Dose intensity is often restricted by severe adverse effects, particularly dermatological, gastrointestinal, ocular, and cardiopulmonary toxic effects. Predictive biomarkers, such as tumour mutational burden, interferon signatures, and MAPK pathway activity, are emerging as crucial tools for refining patient selection and monitoring therapeutic response. Advances in drug design, including dual-targeting strategies, aim to expand the therapeutic window and overcome resistance mechanisms. Combination regimens, particularly those incorporating immune checkpoint inhibitors or PI3K–mTOR pathway inhibition, show promise for enhancing efficacy and treatment durability. Beyond oncology, MEK pathway modulation is under investigation in fibrotic, inflammatory, and developmental disorders, although clinical validation remains at an early stage. Building on more than a decade of use in *BRAF*^{V600} (ie, Val600)-mutant melanoma, MEK inhibitors continue to be refined through biomarker-guided combination strategies and exploration in additional cancers and non-oncological diseases.

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Introduction

The mitogen-activated protein kinase (MAPK) pathway—also known as the RAS–RAF–MEK–ERK cascade—is a highly conserved signalling axis that governs fundamental processes in cellular and organismal development and tissue homoeostasis, including organogenesis, proliferation, survival, and migration.[1,2] Dysregulated MAPK activity is a hallmark of cancer, with activating mutations in core components such as *RAS* (eg, *KRAS* and *NRAS*) and *BRAF,* or loss-of-function mutations in negative regulators, such as *NF1* (which encodes neurofibromin 1), identified in approximately one-third of all human malignancies (figure 1).[3–6] Beyond cancer, germline mutations that affect more than 20 MAPK-related genes, including *PTPN11, SOS1, SOS2, RAF1, KRAS, BRAF, MAP2K1* (which encodes MEK1), and *NF1,* underpin a spectrum of developmental disorders collectively termed RASopathies. These disorders, which include Noonan syndrome, neurofibromatosis type 1, LEOPARD syndrome, and cardiofaciocutaneous syndrome, share phenotypes such as congenital cardiac defects, facial dysmorphism, impaired growth, learning disabilities, and skeletal and ectodermal anomalies.[7]

Given its central role in tumourigenesis and developmental disorders, the MAPK cascade has been a major focus of targeted drug development. Signal transduction proceeds through a canonical three-tiered kinase cascade (RAF–MEK–ERK), with MEK1 and MEK2 functioning as the sole activators of the terminal kinases ERK1 and ERK2.[8,9] This exclusive positioning renders MEK a particularly attractive therapeutic target, and several MEK inhibitors have gained clinical approval. These inhibitors are most effective in combination with BRAF inhibitors, eliciting major, occasionally durable, responses in patients with *BRAF*^{V600} (ie, Val600)-mutant malignancies, and improving patient survival.[10]

The clinical use of MEK inhibition remains constrained by dose-limiting toxic effects, intrinsic and acquired resistance, and adaptive feedback signalling that restores MAPK signalling. These challenges have prompted increasing interest in rational combination strategies aimed at augmenting therapeutic response and circumventing resistance. Strategies under investigation include cotargeting parallel oncogenic pathways, integrating MEK inhibitors with cytotoxic chemotherapy, and combining MEK inhibitors with immune checkpoint inhibitors.[11,12] MEK inhibitors have also been repurposed for the treatment of RASopathies, with some notable

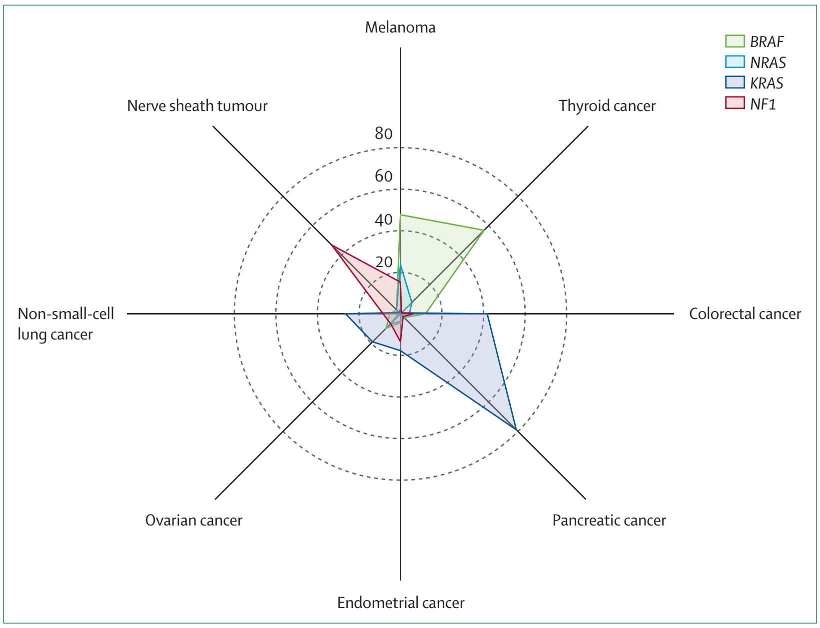

Figure 1: **Comparative frequencies of *BRAF*, *KRAS*, *NRAS*, and *NF1* alterations across tumour subsets**
Radar plot showing the frequency of alterations in *BRAF*, *KRAS*, *NRAS*, and *NF1* across 95 474 tumour samples from the curated The Cancer Genome Atlas Program dataset. Overall, 24% of tumours harboured alterations in these genes, whereas alterations in *MAP2K1* or *MAP2K2* were observed in fewer than 2% of cases. Each axis represents one cancer type, with values indicating the frequency of alterations. Data were obtained from cBioPortal and visualised using Flourish.

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improvements, although long-term safety and optimal dosing remain under investigation.[13]

This Therapeutics paper provides a comprehensive overview of MEK biology, the clinical development of MEK inhibitors, and their therapeutic applications. We discuss key challenges, such as resistance mechanisms, and explore emerging therapeutic strategies and future directions for the clinical application of MEK inhibition in cancer therapy.

The role of MEK in MAPK signalling

The MAPK pathway, anchored by the canonical RAF–MEK–ERK kinase cascade, is initiated by RAS, a small GTPase that cycles between inactive GDP-bound and active GTP-bound states in response to extracellular growth factor signalling (figure 2). Upon ligand binding, receptor tyrosine kinases (RTKs) undergo auto-phosphorylation and recruit the adaptor proteins GRB2 and SOS, which facilitate GDP to GTP exchange on RAS,

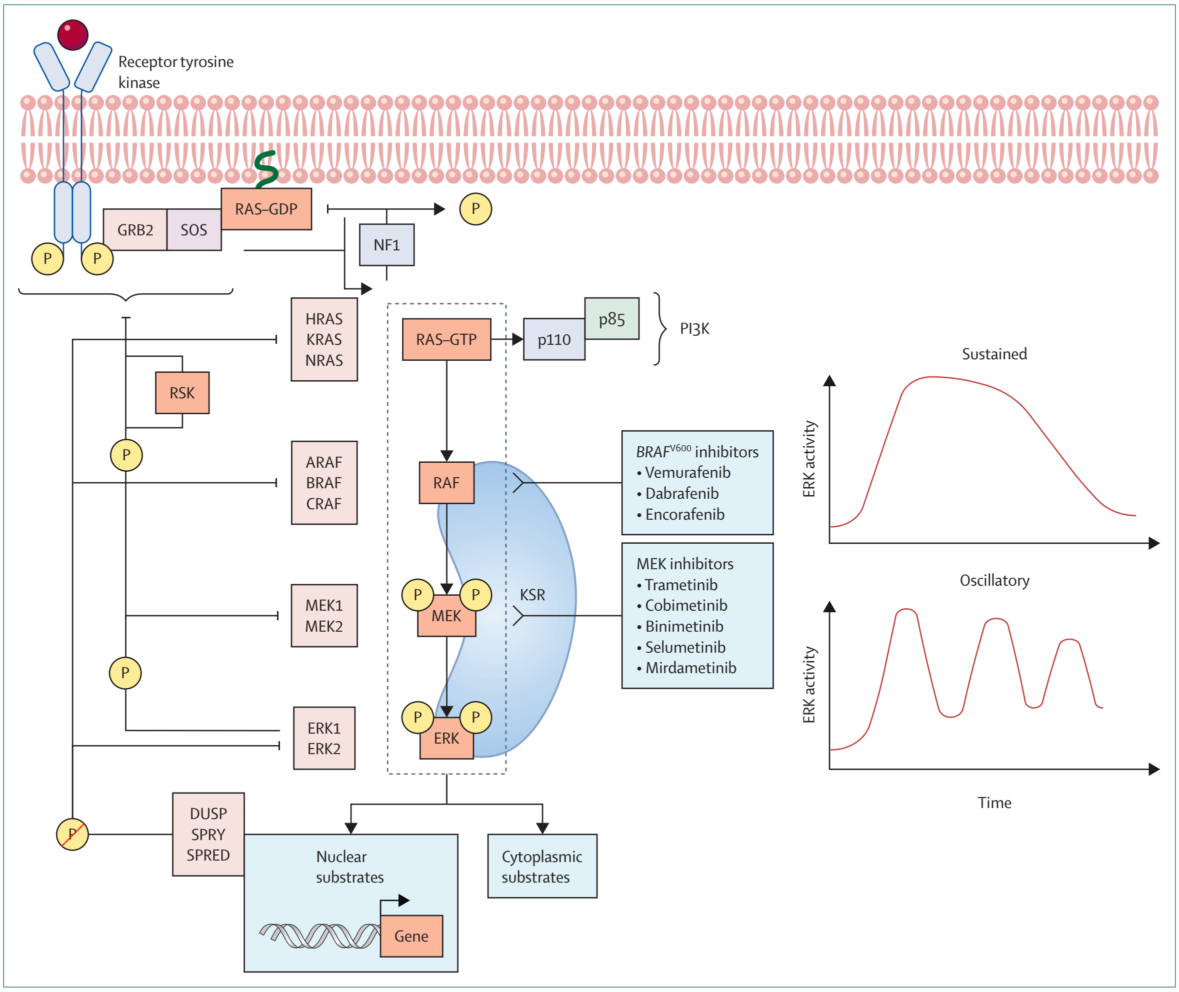

Figure 2: **Overview of canonical MAPK–ERK signalling cascade and regulatory feedback mechanisms**
The dotted boxed region denotes the core MAPK module, highlighting the sequential phosphorylation cascade from RAF to MEK to ERK. Activation of receptor tyrosine kinases at the plasma membrane triggers recruitment of adaptor proteins GRB2 and the guanine nucleotide exchange factor SOS, facilitating the conversion of RAS from its inactive GDP-bound form to the active GTP-bound state. Active RAS interacts with and activates RAF kinases (ARAF, BRAF, and CRAF), which subsequently phosphorylate and activate the dual-specificity kinases MEK1 and MEK2. Activated MEK1 and MEK2 then phosphorylate the extracellular signal-regulated kinases ERK1 and ERK2, which translocate to the nucleus to regulate nuclear substrates, including transcription factors, while also phosphorylating cytoplasmic targets to modulate diverse cellular responses. In parallel, RAS–GTP can also interact with the p110 catalytic subunit of PI3K, activating the PI3K signalling pathway. Feedback regulation is mediated by phosphorylation-dependent inhibitory loops: ERK phosphorylates multiple upstream RTKs, adaptor proteins, BRAF, CRAF, and MEK1, to dampen MAPK signalling, and RSK, a downstream effector of ERK, phosphorylates upstream components, including SOS and CRAF, to attenuate pathway output. Additional negative regulators, including phosphatases DUSPs and the inhibitory SPRY and SPRED proteins, modulate RAS and ERK activity to fine-tune signal intensity and duration. The RAS GTPase-activating protein NF1 promotes RAS inactivation by stimulating hydrolysis of GTP to GDP. The scaffold protein KSR facilitates the spatial organisation of the pathway components to enhance signalling efficiency. Clinically approved BRAF inhibitors and MEK inhibitors are indicated. Temporal dynamics of MAPK are shaped by the interplay of feedback loops, scaffold proteins, and phosphatases. Oscillatory ERK signalling, often driven by pulsatile upstream inputs and rapid feedback, is associated with transient gene expression, whereas sustained ERK activation typically results from persistent upstream stimulation or impaired feedback. Phosphorylation events are indicated by circled P symbols; arrows denote activation, and T-shaped lines represent inhibition. Figure created with BioRender.com. DUSPs=dual-specificity phosphatases.

thereby activating RAS at the plasma membrane. Activated RAS–GTP binds the RAS-binding domain of the RAF family of serine–threonine MAP3Ks (ARAF, BRAF, and CRAF), relieving autoinhibition.[14] RAF isoforms differ in intrinsic kinase activity, with BRAF having the highest activity, followed by CRAF and ARAF.[15,16] Membrane recruitment promotes RAF protein dimerisation, either as homodimers or heterodimers (with BRAF–CRAF predominating in RAS-mediated signalling), which is required for full kinase activation.[14]

Once activated, RAF proteins phosphorylate and activate the dual-specific MAPK kinases MEK1 and MEK2 by targeting serine residues within their activation loop—specifically Ser218 and Ser222 in human MEK1.[17] MEK1 and MEK2 then phosphorylate the terminal MAPKs ERK1 and ERK2 on conserved threonine and tyrosine residues within their activation loop, such as Thr202 and Tyr204 in human ERK1,[18] leading to ERK activation (figure 2). Although MEK1 and MEK2 serve as the exclusive upstream kinases for ERK1 and ERK2, their own phosphorylation and activation can be mediated by a diverse array of MAP3Ks, including mixed-lineage kinases 1–4, MEKK1, MEKK3, PAK1, and COT (figure 3).[28–32]

Activated ERK1 and ERK2 proteins directly regulate more than 600 substrates and indirectly influence approximately 1800 additional targets through diverse mechanisms.[33] These mechanisms include phosphorylation of nuclear transcription factors such as Elk1, c-Fos, and members of the Myc family;[34–36] stabilisation of target mRNA, such as VEGFR, EGFR, DUSP6, and CDKN1A, via phosphorylation of RNA-binding proteins;[37,38] and modulation of chromatin structure and RNA processing.[39] Collectively, ERK-dependent signalling orchestrates global gene expression programmes that govern cell fate decisions, including cell proliferation, differentiation, and survival.

MEK and ERK also participate in tightly regulated negative feedback loops that shape the dynamic behaviour

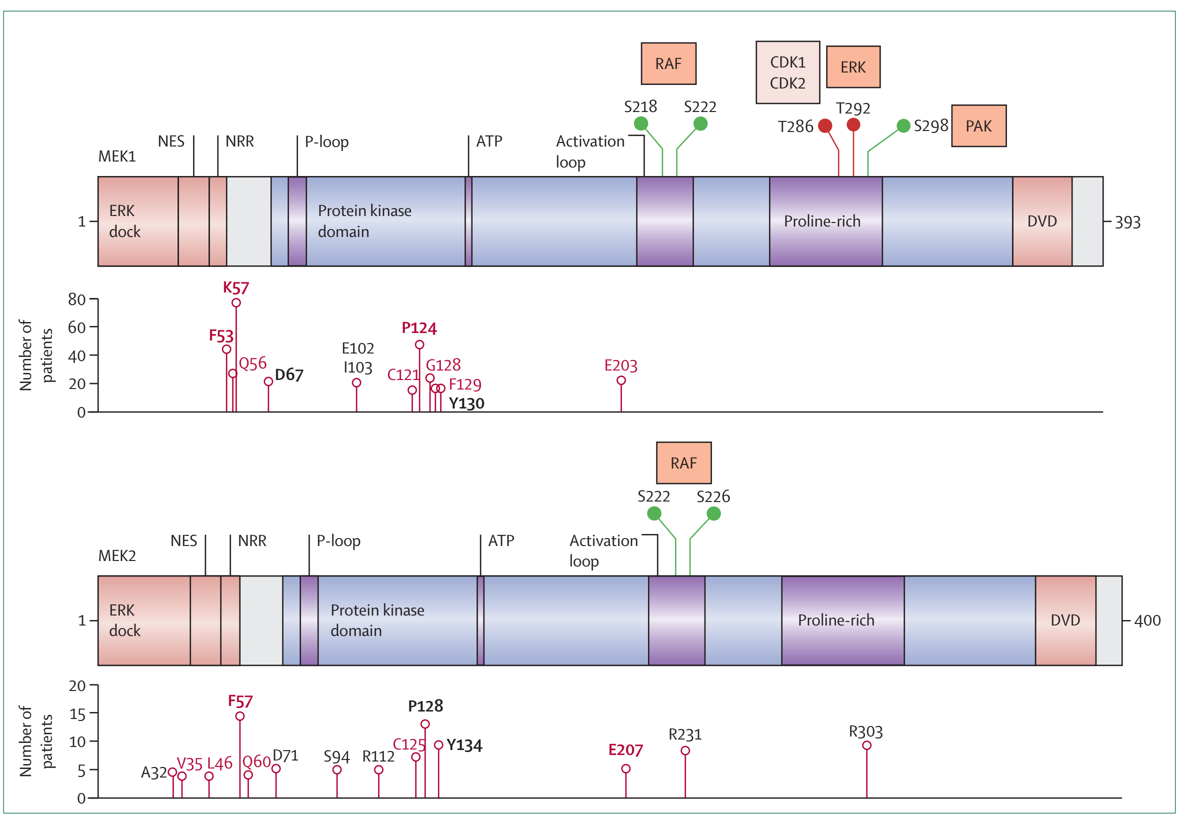

Figure 3: **Human MEK1 and MEK2 protein domains, key regulatory sites, and frequent cancer-associated mutations**
The multifunctional N-terminal region contains the ERK-binding domain (ERK dock), the NES and the NRR. The central kinase domain includes the glycine rich P-loop, the ATP-binding pocket, the activation loop with the RAF (MAP3K)-targeted serine phosphorylation sites, and the proline-rich domain. Key phosphorylation sites are annotated: activating sites are shown in green and inhibitory sites in red, along with the kinases responsible for their modifications. The C-terminal region features an unstructured DVD motif for binding with upstream MAP3Ks.[19] In MEK1, Thr292 is an ERK1-mediated and ERK2-mediated feedback phosphorylation site (absent in MEK2), whereas Ser298 is an activating site phosphorylated by PAK. Below each MEK schema, the frequency and distribution of common cancer-associated mutation sites are shown, and mutation sites linked with BRAF or MEK inhibitor resistance (or both) are in red text.[20–22] Cancer-associated mutation frequency was derived from a curated set of non-redundant studies in cBioPortal (95 474 samples).[23,24] RASopathy MEK1 and MEK2 variants also cluster near the NRR and amino-terminal region of the kinase domain, and common RASopathy variants are in bold text (data were obtained from the Clinical Genome Resource).[25] The schematic design and annotation were informed by literature.[8,26,27] Figure created with BioRender.com. CDK=cyclin-dependent kinase. DVD=domain for versatile docking. NES=nuclear export sequence. NRR=negative regulatory region.

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of MAPK signalling, often producing transient pulses or oscillation patterns of activity rather than sustained activation (figure 2).[40,41] Activated ERK1 and ERK2 phosphorylate multiple upstream components, including RTKs (eg, EGFR, HER2, and ERBB3), adaptor proteins (eg, GAB2, FRS2, IRS1, and IRS2), the RAS guanine nucleotide exchange factor SOS, RAF kinases (eg, BRAF and CRAF), and MEK1,[42–44] to rapidly dampen MAPK intensity and duration.[7,45,46] In addition to this immediate attenuation, ERK activation induces delayed but sustained pathway suppression through transcriptional upregulation of negative regulators, including the SPRY and SPRED proteins and dual-specificity phosphatases (DUSPs). DUSPs selectively dephosphorylate and inactivate ERK, thus terminating MAPK signalling, whereas SPRY and SPRED proteins inhibit RAS and RAF activation (figure 2).[43,47] Downstream ERK effectors, such as the ribosomal S6 kinases, also reinforce negative feedback by phosphorylating and inhibiting upstream components, including SOS and RAF, thereby constraining the amplitude and duration of ERK activity (figure 2).[48] These feedback mechanisms are essential for maintaining MAPK pathway homoeostasis and preventing aberrant cellular proliferation and differentiation.

MAPK signalling is also regulated by the KSR family of scaffold proteins, which function as downstream effectors of RAS. KSR proteins physically interact with all three kinases in the kinase cascade; MEK is constitutively bound, whereas RAF and ERK associate in a stimulus-dependent manner (figure 2). This dynamic scaffolding facilitates the spatial and temporal assembly of kinase complexes, thereby enhancing signal fidelity and propagation.[49] Disruption of these regulatory mechanisms, whether through genetic mutation or dysregulation, contributes to pathological hyperactivation of MAPK signalling, a hallmark of cancer and developmental disorders.

Targeting MEK in cancer: mechanistic and biological rationale

MEK1 and MEK2 share more than 80% of their amino acid identity and possess a highly conserved overall structure, including a flexible N-terminal domain with three key elements: an ERK docking region, a nuclear export signal, and an inhibitory segment that stabilises the kinase in its inactive confirmation. The central protein kinase domain contains the kinase activation loop and a proline-rich domain with multiple activating and inhibitory phosphorylation sites. Activation of MEK requires MAP3K-mediated phosphorylation of two conserved serine residues (Ser218 and Ser222 in human MEK1). The C-terminal sequence of MEK contains a 20 amino acid-docking motif essential for specific interactions with upstream MAP3Ks (figure 3).[19,50–52] Although the MEK1 and MEK2 kinase domains are highly similar, they diverge in their N-terminal and proline-rich domains, which share only 40% of their amino acid identity. These structural differences contribute to distinct biological functions. MEK1, but not MEK2, forms complexes with CRAF and contains a unique phosphorylation site at Thr292 that mediates ERK-dependent feedback, a regulatory feature absent in MEK2. Functional divergence is underscored by genetic studies: MEK2 knock-out mice are viable and phenotypically normal, whereas MEK1 deletion results in embryonic lethality.[53]

To date, the US Food and Drug Administration (FDA) has approved five MEK inhibitors: trametinib, cobimetinib, binimetinib, selumetinib, and mirdametinib. These agents are ATP non-competitive inhibitors that bind to a unique allosteric site adjacent to the ATP site (figure 3). This interaction stabilises MEK1 and MEK2 in their naturally inactive conformation,[54] which results in potent inhibition at nanomolar concentrations (table 1). The high target specificity of these inhibitors[57] is attributed to the complete sequence identity of the allosteric binding sites of MEK1 and MEK2, and the limited sequence and structural conservation of this allosteric pocket across the broader kinome.[54,58]

MEK inhibitors have greatest clinical efficacy in tumours with *BRAF*^{V600} mutations, in which the MAPK pathway is constitutively active and less susceptible to feedback reactivation. In this context, RAS and CRAF activity is typically lower than in RAS-driven or RTK-driven tumours.[59] By contrast, in tumours with *BRAF*^{WT}, non-V600 *BRAF* mutations,[60] or *NF1* loss-of-function mutations,[61] and in tumours driven by oncogenic RAS or activated RTKs, MEK inhibitors show little and variable efficacy. This low sensitivity is largely due to MEK inhibitor-induced relief of ERK-dependent negative feedback on RAF, which promotes RAF dimerisation and activation. The resulting paradoxical, RAF-dependent phosphorylation of MEK1 and MEK2 leads to reactivation of ERK1 and ERK2 signalling[62] and thereby undermines the inhibitory effects of MEK-targeted therapies.

Consequently, complete and durable suppression of MAPK signalling remains challenging in tumours driven by upstream activators such as RAS mutations or RTK activation, and MEK inhibitor monotherapy has not produced meaningful clinical responses in these settings.[63–65] In the NEMO trial, a modest improvement in progression-free survival was observed with the MEK inhibitor binimetinib versus dacarbazine in patients with *NRAS*^{mut} melanoma (median progression-free survival 2·8 months *vs* 1·5 months; hazard ratio [HR] 0·62 [95% CI 0·47–0·80]), although this benefit did not translate into a survival advantage.[66] Even newer-generation MEK inhibitors, such as trametinib and cobimetinib, which not only inhibit the catalytic activity of MEK but also reduce rebound activation by disrupting RAF–MEK complex formation, remain less potent in *BRAF*^{WT} tumours.[62,63] Similarly, in the METRIC trial, the MEK inhibitor trametinib improved progression-free survival compared with dacarbazine chemotherapy

	Trametinib	Cobimetinib	Binimetinib	Selumetinib	Mirdametinib
Biopharmaceutics classification system class	Class II	Class I	Class IV	Class IV	Class I
In vitro enzyme[55,56]	MEK1 IC_{50}=0·7 nM; MEK2 IC_{50}=0·9 nM	MEK1 IC_{50}=0·95 nM; MEK2 IC_{50}=199 nM	MEK1 IC_{50}=12 nM; MEK2 IC_{50}=46 nM	MEK1 IC_{50}=10–14 nM	MEK1 IC_{50}=1 nM; MEK2 IC_{50}=1 nM
Recommended dosage	2 mg once a day	60 mg once a day, for 21 days of a 28-day cycle followed by 7 days of no treatment	45 mg twice a day	25 mg/m^2 twice a day	2 mg/m^2 twice a day, for 21 days of a 28-day cycle followed by 7 days of no treatment
Elimination half-life	4–5 days	43·6 h	3·5 h	6·2 h	28 h
Peak time	1·5 h	2·4 h	1·5 h	1 h	0·8 h
Plasma clearance	5·4 L/h	13·8 L/h	20·2 L/h	8·8 L/h	6·3 L/h
Approved indications	Monotherapy: unresectable or metastatic *BRAF*^{V600E} and *BRAF*^{V600K} melanoma (2013); combination (with dabrafenib): unresectable or metastatic *BRAF*^{V600E} and *BRAF*^{V600K} melanoma (2014), *BRAF*^{V600E} NSCLC (2017), adjuvant *BRAF*^{V600E} and *BRAF*^{V600K}-mutant melanoma (2018), *BRAF*^{V600E} anaplastic thyroid cancer (2018), and *BRAF*^{V600E} solid tumours (2022)	Monotherapy: histiocytic neoplasms (eg, Erdheim–Chester disease and Langerhans cell histiocytosis; 2022); combination (with vemurafenib): unresectable or metastatic *BRAF*^{V600E} and *BRAF*^{V600K} melanoma (2015); triplet therapy (cobimetinib + vemurafenib + atezolizumab): metastatic *BRAF*^{V600E} and *BRAF*^{V600K} melanoma (2020)	Combination (with encorafenib): unresectable or metastatic *BRAF*^{V600E} and *BRAF*^{V600K} melanoma (2018), and *BRAF*^{V600E} metastatic NSCLC (2023)	Monotherapy: paediatric, symptomatic, inoperable neurofibromatosis type 1-related plexiform neurofibromas (2020)	Monotherapy: adult and paediatric patients (≥2 years) with neurofibromatosis type 1 who have symptomatic plexiform neurofibromas not amenable to complete resection (2025)
Common adverse reactions	Fatigue, nausea, rash, diarrhoea, myalgia, peripheral oedema, and acneiform dermatitis	Fatigue, nausea, rash, diarrhoea, photosensitivity, peripheral oedema, and acneiform dermatitis	Fatigue, nausea, rash, diarrhoea, and peripheral oedema	Fatigue, nausea, diarrhoea, oedema, acneiform dermatitis, and asymptomatic creatine kinase elevation	Fatigue, nausea, vomiting, rash, diarrhoea, and creatine kinase elevation

Class I refers to high solubility and high permeability, class II refers to low solubility and high permeability, class III refers to high solubility and low permeability, and class IV refers to low solubility and low permeability. Dabrafenib, vemurafenib, and encorafenib are selective inhibitors of *BRAF*^{V600}-mutant kinases; atezolizumab is an anti-PD-L1 immune checkpoint inhibitor. Approved indications accessed from the US FDA. FDA=Food and Drug Administration. IC_{50}=half maximal inhibitory concentration. NSCLC=non-small-cell lung cancer.

Table 1: **MEK inhibitors approved by the US FDA for clinical use**

(median 4·9 months *vs* 1·5 months; HR 0·54 [95% CI 0·41–0·60]) in patients with *BRAF*^{V600E} (ie, Val600Glu)-mutant or *BRAF*^{V600K} (ie, Val600Lys)-mutant melanoma, establishing MEK inhibition as clinically active but inferior to BRAF-targeted approaches.[67,68]

A notable exception to MEK inhibitor limitations is observed in paediatric patients with neurofibromatosis type 1-associated plexiform neurofibromas, in whom MEK inhibition with selumetinib monotherapy induces clinically meaningful tumour regression, with 17 (71%) of the 24 patients in the trial having confirmed partial response (table 1).[69] Neurofibromatosis type 1 is caused by germline loss-of-function mutations in the *NF1* gene, which encodes a key negative regulator of RAS, and benign tumours develop following somatic loss of the remaining wild-type *NF1* allele. The favourable response to MEK inhibition in these tumours is likely to reflect their genetic stability and a paucity of additional oncogenic mutations beyond this somatic loss.[69,70]

In parallel, additional inhibitors targeting MAPK signalling, including ERK inhibitors, which act downstream of MEK to block ERK-mediated transcription and feedback; SHP2 inhibitors, which prevent RAS activation by RTKs; and SOS1 inhibitors, which disrupt the GRB2–SOS complex, are under investigation. First-in-class ERK inhibitors, such as ulixertinib (BVD-523), have shown on-target ERK suppression and early clinical activity, including in tumours resistant to RAF or MEK inhibitors,[71] although most remain in early-phase trials. SHP2 inhibitors (eg, RMC-4630 and TNO155)[72] and the SOS1 inhibitor BI-1701963[73] are being evaluated both as monotherapy and in combination with other agents, including MEK inhibitors. Several trials with BI-1701963 have been terminated due to toxicity.[73] These approaches highlight ongoing efforts to target multiple nodes of MAPK signalling to improve durable responses in patients with MAPK-driven cancers.

Overcoming BRAF-inhibitor resistance and toxicity: the central role of MEK inhibition

Selective BRAF inhibitors were the first targeted therapies to show a survival benefit in patients with *BRAF*^{V600}-mutant melanoma. FDA-approved selective *BRAF*^{V600} inhibitors, including vemurafenib, dabrafenib, and encorafenib, act by binding the ATP-binding pocket of mutant BRAF proteins, effectively targeting *BRAF*^{V600E}, *BRAF*^{V600D} (ie, Val600Asp), *BRAF*^{V600R} (ie, Val600Arg), and *BRAF*^{V600K} variants.[74–76] In patients with advanced *BRAF*^{V600}-mutant melanoma, these inhibitors significantly improve objective response rates (ORRs), progression-free survival, and overall survival compared with dacarbazine (table 2).[89,90] For example, in the BRIM-3 trial of patients with treatment-naive $BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant melanoma, patients treated with vemurafenib had a median progression-free survival of 6·9 months (95% CI 6·1–7·0), compared with 1·6 months (1·6–2·1) in patients treated with dacarbazine chemotherapy, and a median overall survival of 13·6 months (12·0–15·4) versus 9·7 months (7·9–12·8).[89–91] Notably, vemurafenib is approved only for patients with $BRAF^{V600E}$ mutations. In contrast, dabrafenib has shown clinical activity across a broader spectrum of *BRAF* variants (including $BRAF^{V600E}$,

	n	Study focus	ORR, % (95% CI)	Median progression-free survival, months (95% CI)	Median overall survival, months (95% CI)
METRIC, Flaherty et al (2012)[67,68]					
Trametinib	214	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	40% (33·6–47·1)	4·9 (NR)	15·6 (NR)
Dacarbazine or paclitaxel	108	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	14% (8·0–21·9)	1·5 (NR)	11·3 (NR)
Combi-d, Long et al (2014)[77,78]					
Trametinib + dabrafenib	211	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	69% (62–75)	11·0 (8·0–13·9)	25·1 (19·2 to not reached)
Placebo + dabrafenib	212	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	53% (46–60)	8·8 (5·9–9·3)	18·7 (15·2–23·7)
coBRIM, Ascierto et al (2015)[79,80]					
Cobimetinib + vemurafenib	247	Untreated locally advanced or metastatic	70% (63·5–75·3)	12·3 (9·5–13·4)	22·3 (20·3 to not estimable)
Placebo + vemurafenib	248	$BRAF^{V600}$-mutant melanoma	50% (43·6–56·4)	7·2 (5·6–7·5)	17·4 (15·0–19·8)
NEMO, Dummer et al (2017)[66]					
Binimetinib	269	$NRAS^{mut}$ unresectable or metastatic melanoma	41% (11–20)	3·0 (2·8–4·1)	11·0* (8·9–13·6)
Dacarbazine	133	NRAS^{mut} unresectable or metastatic melanoma	9% (3–13)	1·8 (1·5–2·8)	10·1* (7·0–16·5)
SUMIT, Carvajal et al (2018)[81]					
Selumetinib + dacarbazine	97	Metastatic uveal melanoma	3% (NR)	2·8* (NR)	NR
Placebo + dacarbazine	32	Metastatic uveal melanoma	0% (NR)	1·8* (NR)	NR
BEACON, Kopetz et al (2019)[82]					
Binimetinib + encorafenib + cetuximab	222	$BRAF^{V600E}$-mutant colorectal cancer	26% (18–35)	4·3 (4·1–5·2)	9·0 (8·0–11·4)
Encorafenib + cetuximab	216	$BRAF^{V600E}$-mutant colorectal cancer	20% (13–29)	4·2 (3·7–5·4)	8·4 (7·5–11·0)
Investigator choice	193	$BRAF^{V600E}$-mutant colorectal cancer	2% (0–7)	1·5 (1·5–1·7)	5·4 (4·8–6·6)
Combi-v, Robert et al (2019)[83]					
Trametinib + dabrafenib	352	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	64% (59–61)	11·4 (NR)	NR
Vemurafenib	352	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant metastatic melanoma	51% (46–57)	7·3 (NR)	17·2 (NR)
IMblaze370, Eng et al (2019)[84]					
Cobimetinib + atezolizumab	183	Microsatellite-stable metastatic colorectal cancer	3% (0·9–6·3)	1·91* (1·87–1·97)	8·87* (7·00–10·61)
Atezolizumab	90	Microsatellite-stable metastatic colorectal cancer	2% (0·3–7·8)	1·94* (1·91–2·10)	7·10* (6·05–10·05)
Regorafenib	90	Multikinase inhibitor microsatellite-stable metastatic colorectal cancer	2% (0·3–7·8)	2·00 (1·87–3·61)	8·51 (6·41–10·71)
COLUMBUS, Dummer et al (2020)[85,86]					
Binimetinib + encorafenib	192	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant melanoma	64·1% (56·8–70·8)	14·9 (11·0–20·2)	33·6 (24·4–39·2)
Encorafenib	194	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant melanoma	51·5% (44·3–58·8)	9·6 (7·4–14·8)	23·5 (19·6–33·6)
Vemurafenib	191	$BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant melanoma	40·8% (33·8–48·2)	7·3 (5·6–7·9)	16·9 (14·0–24·5)
IMspire170, Gogas et al (2021)[87]					
Cobimetinib + atezolizumab	222	$BRAF^{V600}$-negative melanoma	26·0% (20·1–32·6)	5·5* (3·8–7·2)	NR
Pembrolizumab	224	$BRAF^{V600}$-negative melanoma	31·6% (25·3–38·4)	5·7* (3·7–9·6)	NR
IMspire150, Ascierto et al (2023)[88]					
Cobimetinib + vemurafenib + atezolizumab	160	$BRAF^{V600E}$-mutant melanoma	67% (61–72)	15·1 (11·4–18·4)	39·0 (29·9 to not estimable)
Cobimetinib + vemurafenib + placebo	170	$BRAF^{V600E}$-mutant melanoma	65% (59–71)	10·6 (9·3–12·7)	25·8 (22·0–34·6)

NR=not reported. ORR=objective response rate. *Did not meet statistical significance.

Table 2: **Selected phase 3 clinical trials evaluating MEK inhibitors (monotherapy and combination)**

$BRAF^{V600K}$, and $BRAF^{K601E}$ [ie, Lys601Glu]), which supports its wider therapeutic applications.[92] Encorafenib has similarly shown efficacy across multiple $BRAF^{V600}$ variants.[85] Despite these improvements, response durability remains low and highly variable. Analysis of 135 metastatic lesions from 24 patients showed that only 71 (53%) had a complete response to BRAF inhibitor monotherapy.[93] These findings highlight that BRAF inhibitor monotherapy is insufficient for durable disease control, which probably reflects rapid adaptive resistance with MAPK pathway reactivation and tumour heterogeneity. These results emphasise the need for more effective therapeutic combination strategies to improve long-term outcomes.[89,94,95]

Resistance to BRAF inhibitors is a major therapeutic challenge in the treatment of $BRAF^{V600}$-mutant cancers. Tumour subclones resistant to BRAF inhibition often pre-exist, particularly within larger metastatic lesions, and expand under the selective pressure of treatment.[93,96–98] Most resistance mechanisms converge on the reactivation of ERK signalling despite BRAF inhibition. These resistance mechanisms include the upregulation of RTKs (eg, IGF1R, EGFR, and PDGFR-ß), activating RAS mutations, BRAF amplification or alternative splicing, and mutations in MEK1 or MEK2.[22,99,100] Importantly, a subset of BRAF inhibitor resistance mechanisms are sensitive to downstream inhibition at the MEK1 or MEK2 nodes or the ERK1 or ERK2 nodes.[101,102] For instance, tumours with *BRAF* splice variants, *BRAF* copy number gains, *MEK1* mutations, *RAS* mutations, or elevated *PDGFR*-β expression are resistant to BRAF inhibitors but have some sensitivity to MEK inhibitors, ERK inhibitors, or combination BRAF–MEK inhibitor therapy.[103,104] The recurrent reactivation of ERK signalling in BRAF inhibitor-resistant tumours thus provides a compelling rationale for MEK inhibition, offering a potential strategy to overcome resistance and enhance clinical outcomes.

Another key limitation of BRAF inhibitor monotherapy is the paradoxical activation of ERK signalling in $BRAF^{WT}$ keratinocytes, which promotes the development of cutaneous squamous cell carcinoma and keratoacanthoma in 20–26% of patients with melanoma treated with vemurafenib[90,105] and 6–11% of patients with melanoma treated with dabrafenib.[92,106,107] Cutaneous squamous cell carcinoma lesions typically arise within 2–4 months of BRAF inhibitor treatment[108] and frequently harbour genetic alterations in RAS GTPases.[109] Paradoxical ERK activation in $BRAF^{WT}$ or RAS^{mut} cells occurs through RAF dimerisation wherein binding of the BRAF inhibitor to one RAF protomer induces the transactivation of the partner protomer,[104,110,111] or by relief of autoinhibitory RAF phosphorylation.[112] This sustained RAF activation maintains ERK signalling despite BRAF inhibition, contributing to the formation of cutaneous squamous cell carcinoma and the progression of RAS^{mut} cancers.

Combined BRAF and MEK inhibition mitigates paradoxical ERK activation.[113,114] A meta-analysis reported the incidence of high-grade cutaneous squamous cell carcinoma to be 11·6% (95% CI 9·8–13·8) in patients treated with single-agent BRAF inhibitors, compared with 2·8% (95% CI 1·9–4·0) in patients receiving combined BRAF–MEK inhibitor therapy.[115] Thus, BRAF–MEK inhibitor combination therapy has become standard of care for $BRAF^{V600}$-mutant melanoma as it delays acquired resistance, increases progression-free survival and overall survival, and reduces the risk of secondary skin neoplasms. Three BRAF–MEK inhibitor combinations have received US FDA approval: dabrafenib plus trametinib, vemurafenib plus cobimetinib, and encorafenib plus binimetinib (table 2).

The phase 3 COMBI-d and COMBI-v trials evaluated first-line treatment with the BRAF inhibitor dabrafenib and the MEK inhibitor trametinib in patients with advanced $BRAF^{V600E}$-mutant or $BRAF^{V600K}$-mutant melanoma.[77,83,113] Both trials showed that the combination of BRAF and MEK inhibitors was associated with superior outcomes compared with BRAF inhibitor monotherapy. Pooled analyses of patients treated with dabrafenib and trametinib from the COMBI-d and COMBI-v trials showed a median progression-free survival of 11·1 months (95% CI 9·5–12·8), median overall survival of 25·9 months (22·6–31·5), and ORR of 68% (64–72).[113] An initial phase 2 randomised study indicated that higher doses of trametinib combined with dabrafenib yielded improved outcomes compared with half the recommended monotherapy dose for both dabrafenib and trametinib, and underscores the importance of optimised dosing in combination regimens.[116,117] Subsequently, the BRAF–MEK inhibitor combinations vemurafenib plus cobimetinib and encorafenib plus binimetinib showed improved efficacy compared with vemurafenib monotherapy and encorafenib monotherapy, respectively.[85,114,118] The phase 3 COLUMBUS trial directly compared encorafenib plus binimetinib with encorafenib or vemurafenib monotherapy in patients with $BRAF^{V600}$-mutant melanoma and found a median progression-free survival of 14·9 months (95% CI 11·0–20·2) with the combination, 9·6 months (7·4–14·8) with encorafenib alone, and 7·3 (5·6–7·9) months with vemurafenib.[85] At 7-year follow-up, median melanoma-specific survival was longest in patients with the combination (36·8 months [27·7–51·5]), followed by encorafenib monotherapy (24·2 months [19·9–35·7]) and vemurafenib monotherapy (19·3 months [14·8–25·9]), and 21·2% of patients remained progression free.[86] Beyond melanoma, the dabrafenib–trametinib combination has shown activity in other $BRAF^{V600}$-mutant cancers, including anaplastic thyroid cancer (ORR 69% [95% CI 41–89]), non-small-cell lung carcinoma (ORR 64% [46–79]), and biliary tract cancer (ORR 47% [31–62]).[119–121]

Early-phase clinical studies of the pan-RAF inhibitor naporafenib in combination with the MEK inhibitor trametinib have shown clinically meaningful activity in patients with *NRAS*[mut] melanoma, with an ORR of 46·7% (95% CI 21·3–73·4) and median progression-free survival of 5·52 months at the recommended dose.[122] Anti-tumour activity of this combination was low in patients with non-small-cell lung cancer (NSCLC), with only two patients with *KRAS*[mut] tumours having a partial response.[123] Similarly, patients receiving naporafenib with anti-PD-1 therapy (spartalizumab) showed little but notable response, including one complete response and three partial responses in *NRAS*[mut] melanoma and *KRAS*[mut] NSCLC expansion cohorts, supporting more investigation of pan-RAF combinations.[124] Other novel pan-RAF dimer inhibitors, such as belvarafenib and brimarafenib, are being explored in combination with MEK inhibitors; in a phase 1b trial, cobimetinib plus belvarafenib showed acceptable tolerability and a 38·5% response rate in patients with *NRAS*[mut] melanoma,[125] whereas the combination of brimarafenib with mirdametinib in patients with advanced solid tumours was limited by tolerability concerns in another trial, which was terminated, highlighting challenges in safely combining these agents (NCT05580770).

Although the combination reduces paradoxical ERK-driven cutaneous toxic effects, some systemic adverse events—including fever, vomiting, and diarrhoea—are more frequent and can be more severe than with single-agent BRAF-inhibitor therapy, reflecting additive effects of both drugs.[77,83,114] Fever is particularly common in patients taking dabrafenib plus trametinib, occurring in 51–53% patients (any grade) compared with 28% of patients treated with dabrafenib monotherapy and 73 (21%) of 349 patients treated with vemurafenib monotherapy.[77,83] Although most adverse events are manageable with dose reductions or treatment interruptions, less than 10% of patients have ongoing toxic effects, most commonly arthralgias and myalgias, with BRAF–MEK inhibitors.[126] Overall, the benefit–risk profile continues to favour combination therapy in patients with *BRAF*^{V600}-mutant melanoma.

Beyond BRAF: emerging combination strategies with MEK inhibitors

Although MEK inhibitors have shown the greatest clinical success in combination with BRAF inhibitors in *BRAF*^{V600}-mutant melanoma, ongoing efforts aim to broaden their utility through rational combination strategies that target parallel signalling pathways or exploit tumour immunogenicity. Given frequent coactivation of the MAPK and PI3K–AKT–mTOR pathways in cancer, and the fact that RAS–GTP also directly activates PI3K (figure 2), concurrent inhibition of both pathways has been explored. In a phase 1b trial, the MEK inhibitor cobimetinib was evaluated in combination with the PI3K inhibitor pictilisib (GDC-0941) in patients with advanced solid tumours.[127] Although pharmacodynamic activity was observed, clinical benefit was limited by dose-limiting toxic effects, including elevated serum lipase and creatinine phosphokinase concentrations.[127] Combinations of MEK inhibitors with dual CDK4 and CDK6 inhibitors, including trametinib plus ribociclib, binimetinib plus ribociclib, and triple BRAF–MEK–CDK4 and CDK6 inhibition therapy with encorafenib, binimetinib, and ribociclib, have also been investigated in patients with *NRAS*[mut] or *BRAF*[mut] melanoma. For example, trametinib combined with the dual CDK4 and CDK6 inhibitor palbociclib has shown synergistic activity in *RAS*[mut] cancers by concurrently targeting mitogenic signalling and cell cycle progression.[128–130] These combinations have shown manageable safety profiles, although toxicity increases with the number of drugs. Early signs of clinical activity have been observed,[131–133] with an ORR of 52·4% reported for the triplet regimen.[131] More studies are required to establish the efficacy of MEK inhibitor-based combination therapies.

Another promising approach in targeting MAPK activation is the RAF–MEK clamp inhibitor avutometinib (VS-6766), which simultaneously inhibits MEK1 and MEK2 kinase activity and prevents compensatory MEK reactivation by RAF. In combination with the FAK inhibitor defactinib, avutometinib has shown durable objective responses (44% in *KRAS*[mut] and 17% in *KRAS*[WT] low-grade serous ovarian cancer[134]), with accelerated US FDA approval granted in May, 2025, for adults with *KRAS*[mut] recurrent low-grade serous ovarian cancer who have received previous therapy. This approval marks a potential new standard for a rare tumour type with historically few treatment options.

Beyond targeted therapy, MEK inhibition has also been explored in immunotherapy combinations. Preclinical studies suggest that MEK inhibitors can increase tumour immunogenicity by upregulating major histocompatibility complex class I expression, reducing immunosuppressive cytokine production, and promoting $CD8^+$ T-cell infiltration in the tumour microenvironment.[135–137] These observations provided a rationale for combining MEK inhibitors with immune checkpoint inhibitors. In a multicohort phase 1b study of cobimetinib plus the anti-PD-L1 antibody atezolizumab, 152 patients with advanced solid tumours, including colorectal cancer, melanoma, and NSCLC were treated.[138] Confirmed responses were observed in nine (41%) of 22 patients with melanoma, five (18%) of 28 patients with NSCLC, and three (19%) of 16 patients with other tumours (including ovarian cancer, clear-cell sarcoma, and renal cell carcinoma).[138] Activity was noted even in microsatellite-stable colorectal cancer, which is typically refractory to immunotherapy, suggesting that MEK inhibition might sensitise otherwise immunologically cold tumours to immunotherapy.[138–140]

Building on these findings, the triplet combination of the MEK inhibitor cobimetinib, the BRAF inhibitor vemurafenib, and the PD-L1 inhibitor atezolizumab was evaluated in the phase 3 IMspire150 trial in treatment-naive patients with $BRAF^{V600}$-mutant melanoma.[88,141] The median progression-free survival was 15·1 months (95% CI 11·4 to 18·4) in the atezolizumab group versus 10·6 months (9·3 to 12·7) in the control group, and the median overall survival was 39·0 months (29·9 to not estimable) in the atezolizumab group versus 25·8 months (22·0 to 34·6) in the control group, which led to US FDA approval in July, 2020.[88,141] However, statistically significant efficacy of this strategy was not shown in two other phase 3 melanoma trials (KEYNOTE-022 and COMBI-i) and has not been replicated across other tumour types.[142,143] In the IMblaze370 phase 3 clinical trial, the combination of cobimetinib plus atezolizumab was investigated in patients with previously treated microsatellite-stable colorectal cancer, but did not show a statistically significant overall survival benefit over regorafenib, a small-molecule multikinase inhibitor.[84,139] Similarly, binimetinib has been explored in combination with immune checkpoint inhibitors, including pembrolizumab, nivolumab, and ipilimumab, in patients with advanced solid tumours, such as melanoma and colorectal cancer, reflecting continued efforts to identify synergistic interactions, optimise combination strategies, and improve biomarker-driven patient selection.[144–146]

In addition to immunotherapy, MEK inhibitors are being tested in combination with other therapeutic classes, such as antiangiogenic and apoptotic agents. Multiple clinical trials are underway to assess the safety and efficacy of these combinations across diverse tumour types (table 3).

MEK pathway modulation beyond oncology

Although MEK inhibitors were originally developed and used for oncological indications, evidence increasingly suggests that aberrant MAPK signalling contributes to the pathogenesis of several non-malignant conditions. As a result, MEK inhibitors are being repurposed or investigated for use in a range of inflammatory, fibrotic, and developmental disorders driven by dysregulated RAS–RAF–MEK–ERK signalling.[161]

RASopathies have been among the first non-cancer indications to benefit from targeted MEK inhibition.[162–165] In April, 2020, selumetinib became the first MEK inhibitor approved outside of oncology, receiving US FDA approval for paediatric patients with symptomatic neurofibromatosis type 1-associated plexiform neurofibromas. This approval was based on the phase 2 SPRINT trial, which showed a partial response in 34 (68%) of 50 patients alongside improvement in pain and functional outcomes.[69,166] Beyond neurofibromatosis type 1, case reports also describe trametinib-mediated reversal of severe hypertrophic cardiomyopathy and resolution of chylous effusions in children with Noonan syndrome (caused by germline mutations in genes such as *PTPN11*, *SOS*, and *RAF1*), illustrating the potential for broader applications across RASopathy phenotypes.[167–169]

	Drug action	Study focus
Sullivan et al (2015),[130] phase 1/2, completed		
Trametinib	MEK inhibition	Advanced solid tumours
Palbociclib	CDK4 and CDK6 inhibition	Advanced solid tumours
Carter et al (2016),[147] phase 2, completed		
Selumetinib	MEK inhibition	Metastatic pancreatic adenocarcinoma
Erlotinib	EGFR inhibition	Metastatic pancreatic adenocarcinoma
Lee et al (2016),[128] phase 1, completed		
Binimetinib	MEK inhibition	$KRAS^{mut}$ and $NRAS^{mut}$ colorectal cancer
Palbociclib	CDK4 and CDK6 inhibition	$KRAS^{mut}$ and $NRAS^{mut}$ colorectal cancer
Ascierto et al (2017),[131] phase 1, completed		
Binimetinib	MEK inhibition	Advanced solid tumours
Encorafenib	BRAF inhibition	Advanced solid tumours
Ribociclib	CDK4 and CDK6 inhibition	Advanced solid tumours
Algazi et al (2018),[148] phase 2, completed		
Trametinib	MEK inhibition	$BRAF^{WT}$ and $NRAS^{WT}$ wild-type melanoma
GSK2141795	AKT inhibition	$BRAF^{WT}$ and $NRAS^{WT}$ wild-type melanoma
Bendell et al (2020),[149] phase 1/2, completed		
Cobimetinib	MEK inhibition	$EGFR^{mut}$ NSCLC
Osimertinib	EGFR inhibition	$EGFR^{mut}$ NSCLC
RMC-4630	SHP2 inhibition	$EGFR^{mut}$ NSCLC
Gaudreau et al (2020),[150] phase 1/2, completed		
Selumetinib	MEK inhibition	NSCLC
Durvalumab	Anti-PD-L1 immune checkpoint inhibition	NSCLC
Tremelimumab	Anti-CTLA-4 immune checkpoint inhibition	NSCLC
LoRusso et al (2020),[132] phase 1, terminated per sponsor decision		
Trametinib	MEK inhibition	Advanced solid tumours
Ribociclib	CDK4 and CDK6 inhibition	Advanced solid tumours
Schuler et al (2022),[133] phase 1/2, completed		
Binimetinib	MEK inhibition	$NRAS^{mut}$ melanoma
Ribociclib	CDK4 and CDK6 inhibition	$NRAS^{mut}$ melanoma
Buchbinder et al (2023),[151] phase 1/2, completed		
Dabrafenib	BRAF inhibition	Melanoma post-BRAF– MEK inhibitor progression
Trametinib	MEK inhibition	Melanoma post-BRAF– MEK inhibitor progression
MCS110	CSF-1 inhibition	Melanoma post-BRAF– MEK inhibitor progression
Manji et al (2023),[152] phase 1/2, terminated due to toxicity and lack of efficacy		
Cobimetinib	MEK inhibition	$KRAS^{mut}$ malignancies
Atezolizumab	Anti-PD-L1 immune checkpoint inhibition	$KRAS^{mut}$ malignancies
Hydroxychloroquine	Autophagy inhibition	$KRAS^{mut}$ malignancies
Schjesvold et al (2023),[153] phase 1/2, completed		
Cobimetinib	MEK inhibition	Multiple myeloma
Atezolizumab	Anti-PD-L1 immune checkpoint inhibition	Multiple myeloma
Venetoclax	BCL-2 inhibition	Multiple myeloma

(Table 3 continues on next page)

	Drug action	Study focus
(Continued from previous page)		
Tian et al (2023),[154] phase 2, completed		
Dabrafenib	BRAF inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Trametinib	MEK inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Spartalizumab	Anti-PD-1 immune checkpoint inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Van Cutsem et al (2023),[155] phase 2, completed		
Binimetinib	MEK inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Encorafenib	BRAF inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Cetuximab	EGFR inhibition	*BRAF*^{V600E}-mutant colorectal cancer
Manoharan et al (2024),[156] phase 2, completed		
Cobimetinib	MEK inhibition	Platinum-resistant high-grade serous ovarian cancer
Atezolizumab	Anti-PD-L1 immune checkpoint inhibition	Platinum-resistant high-grade serous ovarian cancer
Bevacizumab	VEGF inhibition	Platinum-resistant high-grade serous ovarian cancer
Mutch et al (2024),[157] phase 1, completed		
Cobimetinib	MEK inhibition	*BRCA*^{WT} ovarian cancer
Atezolizumab	Anti-PD-L1 immune checkpoint inhibition	*BRCA*^{WT} ovarian cancer
Niraparib	PARP inhibition	*BRCA*^{WT} ovarian cancer
Somaiah et al (2024),[158] phase 2, completed		
Selumetinib	MEK inhibition	Malignant peripheral nerve sheath tumours
Sirolimus	mTOR inhibition	Malignant peripheral nerve sheath tumours
Dagogo-Jack et al (2025),[159] phase 1/2, active, not recruiting		
Cobimetinib	MEK inhibition	ALK-rearranged NSCLC
Alectinib	ALK inhibition	ALK-rearranged NSCLC
Prasath et al (2025),[160] phase 2, completed		
Trametinib	MEK inhibition	Triple negative breast cancer
GSK2141795	AKT inhibition	Triple negative breast cancer
NCT03272464, phase 1, terminated due to slow accrual		
Dabrafenib	BRAF inhibition	*BRAF*^{V600E}-mutant or *BRAF*^{V600K}-mutant melanoma and solid tumours
Trametinib	MEK inhibition	*BRAF*^{V600E}-mutant or *BRAF*^{V600K}-mutant melanoma and solid tumours
Itacitinib	JAK1 inhibition	*BRAF*^{V600E}-mutant or *BRAF*^{V600K}-mutant melanoma and solid tumours
NCT03631953, phase 1, recruiting		
Trametinib	MEK inhibition	Refractory meningiomas
Alpelisib	PIK3CA inhibition	Refractory meningiomas
NCT03947385, phase 1/2, recruiting		
Binimetinib	MEK inhibition	GNAQ, GNA11, or PRKC fusion solid tumours
Darovasertib	PKC inhibition	GNAQ, GNA11, or PRKC fusion solid tumours
NCT03979651, phase 1/2, completed		
Trametinib	MEK inhibition	*NRAS*^{mut} melanoma
Hydroxychloroquine	Autophagy inhibition	*NRAS*^{mut} melanoma

(Table 3 continues on next page)

Somatic activation of mutations in MEK pathway genes, particularly *KRAS* and *MAP2K1*, have also been identified in arteriovenous malformations.[170,171] Preclinical studies implicate excessive MEK signalling in endothelial dysregulation and abnormal vessel morphogenesis. Although clinical experience remains rare, trametinib has shown promise in inducing regression and symptomatic improvement of extracranial arteriovenous malformations and in managing capillary malformation–arteriovenous malformation syndrome with cardiac compromise.[172,173]

Fibrotic diseases represent another promising avenue for MEK inhibitor repurposing. Aberrant MEK signalling promotes fibroblast proliferation, extracellular matrix deposition, and production of inflammatory mediators, which are key processes in the development of fibrotic diseases.[174] In preclinical models, MEK inhibition with PD 0325901 and trametinib reduced fibrosis in lung and liver injury.[175,176] Clinical trials are in early stages, but MEK inhibitors are under investigation in idiopathic pulmonary fibrosis and systemic sclerosis.[174,177]

Hyperactivation of MAPK signalling has been implicated in the pathogenesis of several autoimmune and inflammatory diseases, including rheumatoid arthritis, psoriasis, and inflammatory bowel disease.[178–181] MEK inhibition has been shown to modulate cytokine production, reduce T-cell activation, and impair pro-inflammatory signalling in preclinical models;[135] however, the systemic inhibition of MEK in chronic inflammatory settings poses substantial toxicity concerns, and no MEK inhibitor has yet been approved for these indications.

In gynaecological disorders, deep infiltrating endometriosis, a benign yet frequently debilitating condition, has been shown to harbour somatic mutations in oncogenic drivers, such as *KRAS* and *PIK3CA*.[182] These genetic alterations converge on the PI3K–AKT–mTOR and MEK–ERK signalling pathways, implicating aberrant pathway activation in both lesion proliferation and chronic inflammation. Preclinical data suggest that targeted inhibition of these pathways might attenuate disease progression and symptom burden; however, clinical studies are scarce.[183–185]

Challenges and limitations of MEK inhibitor therapy

Drug resistance

Despite early promise, MEK inhibitors have shown little and often non-durable clinical activity across most tumour types. Whereas combination strategies, particularly with BRAF inhibitors, offer improved outcomes in some populations, dual BRAF–MEK inhibition rarely results in sustained disease control. Resistance is almost inevitable, as single genetic or signalling alterations can bypass combination pathway inhibition.[96] Moreover, MEK inhibitor sensitivity is shaped by adaptive signalling, tumour microenvironmental cues, and epigenetic reprogramming, all of which contribute to treatment failure.

MAPK pathway reactivation in the presence of BRAF inhibitors, MEK inhibitors, or both, frequently arises through alterations that directly affect drug targets, such as *BRAF* amplification or mutations in *MEK1* and *MEK2*.[98,186–189] Notably, resistance to dabrafenib and trametinib in patients with melanoma occurs predominantly through mutations in *MEK2* rather than *MEK1*,[20] potentially reflecting ERK-mediated inhibition of MEK1 through phosphorylation at the MEK1-specific regulatory residue Thr292 (figure 2). Phosphorylated MEK proteins also show reduced affinity for trametinib, diminishing drug efficacy.[190]

Upstream events, including *RAS* mutations and upregulation of RTKs such as EGFR and HER2, can co-activate MAPK and compensatory PI3K–AKT signalling, enabling tumour cells to bypass BRAF and MEK inhibition.[191,192] Activation of PI3K–AKT signalling can also occur via oncogenic mutations in *PIK3CA, AKT1*, or *AKT3*, or via loss of the tumour suppressor *PTEN*.[193,194] In gastric cancer, for instance, MEK blockade induces rebound activation of $KRAS^{WT}$ and the SHP2 phosphatase, promoting RTK-dependent signalling and resistance.[195] MEK inhibition relieves ERK-mediated negative feedback on RTKs, such as EGFR and HER2, thereby enhancing ERBB3-driven PI3K–AKT signalling and contributing to resistance.[196]

In colorectal cancer, MEK inhibition persistently reduced AXIN1 protein concentrations and disrupted its interaction with GSK3B, leading to activation of Wnt, stem-cell plasticity, and therapy resistance.[197] Similarly, MEK inhibition can activate STAT3 via FGFR and JAK kinases, creating a positive feedback loop that sustains survival signalling in drug-treated, oncogene-addicted cancer cells.[198]

These mechanistic insights suggest rational avenues for combination therapies aimed at delaying or overcoming MEK inhibitor resistance. Cotargeting MEK and RTKs (eg, EGFR, HER2, or ERBB3) or inhibiting parallel survival pathways, including PI3K–AKT or Wnt signalling, has shown promise in preclinical models. Dual inhibition strategies, such as combining MEK with SHP2 or PI3K inhibitors,[199] are being explored to suppress adaptive feedback loops and restore drug sensitivity. These findings underscore the potential of context-specific, mechanism-driven combinations to improve the durability of MAPK pathway inhibition across diverse tumour types.

The tumour microenvironment can also contribute to MEK inhibitor resistance. HGF and FGF secreted by stromal cells (eg, fibroblasts) in the tumour microenvironment can reactivate MAPK signalling or stimulate alternative survival pathways. For example, HGF binding to its receptor MET activates the PI3K–AKT pathway and promotes cell survival, despite MEK suppression.[200] Longitudinal studies on trametinib-treated murine melanoma have shown intratumoural reorganisation involving bundled collagen deposition, which might promote resistance to targeted therapy over time.[201]

	Drug action	Study focus
(Continued from previous page)		
NCT04109456, phase 1, active, not recruiting		
Cobimetinib	MEK inhibition	Metastatic melanoma
IN10018 (FAK)	FAK inhibition	Metastatic melanoma
NCT04201457, phase 1/2, active, not recruiting		
Dabrafenib	BRAF inhibition	Recurrent low-grade or high-grade glioma with BRAF alteration
Trametinib	MEK inhibition	Recurrent low-grade or high-grade glioma with BRAF alteration
Hydroxychloroquine	Autophagy inhibition	Recurrent low-grade or high-grade glioma with BRAF alteration
NCT05585320, phase 1/2, active, not recruiting		
Atebimetinib (IMM-1-104)	MEK inhibition	Metastatic pancreatic ductal adenocarcinoma
Modified gemcitabine and nab-paclitaxel	Inhibition of DNA synthesis and microtubule function	Metastatic pancreatic ductal adenocarcinoma
Dabrafenib	BRAF inhibition	$BRAF^{mut}$ melanoma post anti-PD-(L)1 monoclonal antibody progression

NSCLC=non-small-cell lung cancer.

Table 3: **Interventions in clinical trials involving MEK inhibitor combination therapies**

Toxicity

Dosing of MEK inhibitors is limited by their toxicity profiles, with adverse effects occurring in the majority of treated patients and affecting multiple organ systems. These toxic effects range from mild to severe.[202] Dermatological toxic effects are the most frequent and are often dose-limiting. Rashes, such as acneiform dermatitis and maculopapular eruptions, affect a large proportion of patients and substantially affect their quality of life.[203,204] Gastrointestinal adverse effects, including diarrhoea, nausea, and vomiting, are the second most common and affect more than 20% of patients treated with MEK inhibitors.[202,203] Ocular toxic effects are uniquely associated with MEK inhibitors and include central serous retinopathy and retinal vein occlusion.[203] Whereas serous retinopathy typically resolves with treatment interruption, retinal vein occlusion can result in irreversible vision loss.[204] Cardiotoxicity, such as reduced ejection fraction or ventricular dysfunction, has been reported in up to 7% of patients, and interstitial lung disease or pneumonitis, although less frequent, can be fatal.[203,204] Other common toxic effects include fatigue and peripheral oedema, which compound treatment burden.[205]

These toxic effects not only impair quality of life but also limit dose intensity and duration of therapy. Among MEK inhibitors, induced grade 3–4 events or death were higher among patients prescribed binimetinib (69% [95% CI 50–84]) than for patients prescribed trametinib (36% [17–60]).[202] To improve tolerability while preserving efficacy, several strategies have been explored, including alternative dosing schedules, dose reductions, and lower-dose combination regimens.[206] In a phase 2 trial of trametinib for $BRAF^{V600}$ and $NRAS^{Q61}$ wild-type immune checkpoint inhibitor-refractory melanoma, the combination of trametinib with low-dose dabrafenib (50 mg twice daily) mitigated treatment-limiting skin toxicity while maintaining clinical activity, with a 29% ORR (seven of 24 patients) and disease control in nine (64%) of 14 patients with MAPK pathway-activating alterations.[207]

Future directions and perspectives

MEK inhibitors have emerged as key agents in the treatment of BRAF-driven and RAS-driven malignancies. However, their broader clinical effect has been limited by adaptive resistance, dose-limiting toxic effects, and poor durability of response when used as monotherapy. To fully realise the therapeutic potential of MEK pathway inhibition, several crucial avenues of research must be pursued.

Refining patient selection through predictive biomarkers is essential. Although activating mutations in *RAS* and *RAF* are often associated with MEK inhibitor sensitivity, clinical outcomes remain heterogenous and variable. In the COMBI-AD trial, patients with low tumour mutational burden derived greatest benefit from combination therapy, while those with high tumour mutational burden and low type II interferon signatures showed little response.[208] Additionally, patients with $BRAF^{V600K}$ mutations did not have the same overall survival benefit as those with $BRAF^{V600E}$ mutations, which highlights the need to account for mutational subtype when considering treatment strategies.[209] Whole-exome and RNA sequencing analyses of patients who had a response versus those who did not showed higher rates of *MITF* amplification and *TP53* mutation in non-responding tumours, whereas *NF1* deletion was more common in responding tumours, and transcriptional signatures of CD8 T effector cells, antigen presentation, and natural killer cells were enriched in responders.[210] Emerging tools, such as comprehensive genomic profiling, circulating tumour DNA, and functional assays of MAPK pathway activity offer promise for identifying responsive subgroups and detecting early resistance. Integrating somatic and germline alterations, transcriptional signatures, and pathway activity metrics will be central to identifying durable responders.

Refinements in drug design are expected to improve both efficacy and tolerability. Next-generation MEK inhibitors with enhanced potency and selectivity, reduced toxicity, and alternative formulations (eg, topical or targeted delivery) are in early development.[211] Tunlametinib (HL-085) has shown higher inhibitory activity than some earlier MEK inhibitors in preclinical studies (with significantly lower half maximal inhibitory concentration values and strong suppression of ERK signalling) and encouraging clinical activity in $NRAS^{mut}$ melanoma (ORR 35·8% [95% CI 26·2–46·3]), which has led to regulatory approval in China and ongoing clinical evaluation of combinations with BRAF inhibitors.[212,213] Mechanistically novel compounds, including allosteric variants with alternative binding modes, and dual inhibitors that simultaneously target MEK and upstream or downstream MAPK effectors, could overcome resistance and expand the therapeutic window.

Combination therapy remains a cornerstone of MEK inhibitor development. Dual MAPK pathway inhibition, exemplified by BRAF and MEK inhibitor combinations, has improved survival in patients with melanoma and patients with other cancers. This paradigm is evolving toward triplet regimens incorporating immune checkpoint inhibitors, PI3K–mTOR pathway agents, or proapoptotic drugs. Rational combinations tailored to tumour-specific vulnerabilities guided by genomic and functional profiling are likely to define the next generation of MEK-based therapies. Many current-generation MEK inhibitor-based combinations, including those with immune checkpoint inhibitors, PI3K–mTOR inhibitors, or pan-RAF inhibitors, have shown little activity or unacceptable toxicity. To overcome these challenges, strategies under investigation include lower-dose or intermittent scheduling, sequential rather than concurrent administration, and alternative tumour-directed delivery strategies, such as nanoparticles, liposomes, or RNA-based therapeutics,[214] which aim to improve efficacy and reduce systemic toxicity.

Although oncology remains the primary focus, aberrant MAPK signalling has been implicated in fibrotic, inflammatory, and developmental diseases. Preclinical data suggest potential applications of MEK inhibition beyond cancer, although early-phase clinical evidence is scarce. Translational studies are needed to evaluate safety, efficacy, and therapeutic windows in non-malignant contexts.

The future of MEK inhibitors lies not in monotherapy, but in the integration of MEK inhibitors into biomarker-driven, combination-based, and precision-guided strategies. With ongoing innovations in drug design and a deepening understanding of resistance biology, MEK inhibition is poised to remain a foundational element of targeted therapy and potentially extend its relevance beyond oncology.

Contributors

WYC, IPdS, and HR drafted the initial version of the manuscript. All authors contributed to data analysis and interpretation, critically reviewed and edited subsequent drafts, and approved the final version for submission.

Declaration of interests

GVL is a consultant advisor for Agenus, AstraZeneca UK, Bayer HealthCare Pharmaceuticals, BioNTech, Boehringer Ingelheim International, Bristol Myers Squibb, Byondis, Evaxion Biotech, Fortiva Biologics USA, GI Innovation, Hexal (a Sandoz Company), Highlight Therapeutics, Immunocore Ireland, Innovent Biologics USA, IOBiotech, Iovance Biotherapeutics, Merck Sharp & Dohme, Novartis Pharma, Pierre Fabre, Regeneron Pharmaceuticals, Scancell, SkylineDX, and Strand Therapeutics. IPdS reports travel support from Bristol Myers Squibb and Merck Sharp & Dohme; reports speakers fees from Pierre Fabre, Roche, Bristol Myers Squibb, Merck Sharp & Dohme, and

Novartis; and has served as a consultant on advisory boards from Merck Sharp & Dohme, Regeneron, and Strand. WYC reports a speakers fee from AstraZeneca UK, and is supported by the CLEARbridge Foundation, a Melanoma Institute Australia PhD scholarship, and a Tour de Cure Postgraduate Research Grant. HR declares no competing interests.

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